SUBMISSION-QUALITY KNOWLEDGE HUB

Common Bioequivalence Study-Design Deficiencies

A quick reference for avoiding first-cycle BE deficiencies in ANDAs

Prepared by Sumiya Mushtaq, PharmD, MBA

Version 1.0 · Content current as of June 2026. FDA guidance changes — verify against the latest sources before relying on this guide.

Independent educational resource — please read

This guide is an independent, educational resource prepared from publicly available FDA guidance and the author's professional experience. It is not affiliated with, authored by, endorsed by, or sponsored by the U.S. Food and Drug Administration or any other government agency.

It reflects a manufacturer / applicant-side perspective on submission quality. It is general information, not regulatory, legal, or clinical advice, and does not replace the current FDA guidance documents, the Code of Federal Regulations, or consultation with qualified regulatory professionals. Always verify against the latest FDA sources before acting.

Why first-cycle BE quality matters

Bioequivalence deficiencies are among the most common reasons ANDAs do not clear the first review cycle, and they recur predictably across applications. FDA reviewers have repeatedly shown that most BE deficiencies are avoidable. The single most-cited root cause of BE-related complete response letters is deviating from a product-specific guidance (PSG) without prior FDA agreement.

The one rule that prevents the most CRLs

Follow the product-specific guidance (PSG). If you must deviate — on study design, dissolution method, or endpoints — obtain FDA agreement first through a Controlled / Control Correspondence rather than discovering the disagreement at review.

Where the deficiencies actually fall

In a published FDA review of first-cycle BE assessments for extended-release (ER) oral tablet products, 224 deficiencies were identified across 157 ANDAs. The distribution shows where attention pays off:

Deficiency area	Share	Most common sub-issues
Dissolution	37%	Missing/inadequate multi-pH media; inadequate data for scored tablets; non-discriminating method; missing alcohol dose-dumping data.
Bioanalysis	35%	Missing raw data/chromatograms; inadequate sample re-analysis; missing SOP/validation; missing/inadequate stability data.
Pharmacokinetics	17%	Subject exclusion without justification; inadequate info on failed/pilot study; incomplete SAS transport files.

Deficiency area	Share	Most common sub-issues
Formulation	11%	Excipient level exceeded; missing data on colorant, flavor, or capsule shell.

Note: these percentages are for ER oral tablet products in the cited FDA analysis. The categories generalize well, but always check the PSG for your specific product.

Quick-reference: deficiency → prevention

Dissolution

- **Use the PSG.** Confirm the recommended apparatus, media, and conditions before generating data.
- **Show discrimination.** Demonstrate the method can distinguish meaningful formulation/process changes.
- **Cover multi-pH media** where recommended; don't submit single-medium data when multiple are expected.
- **Scored tablets:** provide whole- and half-tablet comparative data; understand the RLD's formulation design before splitting.
- **Test on a fresh lot** or confirm test-product stability at the time of the dissolution study.
- **Include alcohol dose-dumping data** for ER products where applicable.

Bioanalysis

- **Submit 100% of analytical raw data** including rejected and repeat runs, plus chromatograms from ~20% of serially selected subjects.
- **Justify every re-analysis** with supporting documentation and data.
- **Include the SOP and full method validation,** and adequate analyte stability data.

Pharmacokinetics & statistics

- **Justify subject exclusions** with a real-time investigation report and supporting evidence — never post-hoc removal to pass.
- **Report failed and pilot studies** adequately; don't omit them.
- **Provide CDISC/SDTM-compliant data** (required for clinical trials starting after 17 Dec 2016) and make sure SAS transport files are complete and match the study report.

Formulation

- **Stay within excipient limits;** if unsure whether proposed inactive-ingredient levels are acceptable, file a Controlled Correspondence before submitting.
- **Give a full quantitative breakdown** of colorants, inks, flavors, and capsule shells.

Standard PK BE study design (typical default)

For an orally administered, immediate-release product with a fasting in vivo BE study, FDA generally also recommends a fed study — except where the RLD labeling states the product should be taken only on an empty stomach. The fed study is usually a single-dose, two-period, two-treatment, two-sequence crossover design. Always confirm against the PSG and current general BE guidance.

Guidance update — current as of June 2026

On May 29, 2026, FDA finalized its guidance "Bioequivalence Studies With Pharmacokinetic Endpoints for Drugs Submitted Under an ANDA," and issued a companion final guidance, "Statistical Approaches to Establishing Bioequivalence." For immediate-release solid oral dosage forms, BE study-design recommendations are now harmonized with ICH M13A (issued Oct 31, 2024). The reference-scaled average BE statistical methods for narrow-therapeutic-index and highly variable drugs now live in the separate Statistical Approaches guidance. The general principles in this quick reference still hold, but confirm specifics against these current final guidances and the relevant PSG before designing a study.

Pre-submission BE self-check

PSG located and followed (or deviation pre-cleared via Controlled Correspondence).

Fasting study — and fed study where required — designed and reported.

Dissolution: correct media/apparatus, discriminating, scored-tablet and dose-dumping data where applicable.

Bioanalysis: full raw data, chromatograms, validation, stability, justified re-analyses.

PK/stats: justified exclusions, CDISC-compliant data, matching SAS transport files.

Formulation: within excipient limits, full quantitative breakdown provided.

Sources & further reading

This guide draws on the following publicly available FDA materials and regulations. Always confirm you are using the current version.

- Common Bioequivalence Deficiencies in ANDAs for Extended-Release Drug Products (FDA, OGD, April 2025) — https://www.fda.gov/media/187317/download
- Bioequivalence Studies With Pharmacokinetic Endpoints for Drugs Submitted Under an ANDA (FDA final guidance, May 29, 2026) — https://www.fda.gov/regulatory-information/search-fda-guidance-documents/bioequivalence-studies-pharmacokinetic-endpoints-drugs-submitted-under-abbreviated-new-drug
- Statistical Approaches to Establishing Bioequivalence (FDA final guidance, May 29, 2026) — https://www.federalregister.gov/documents/2026/05/29/2026-10705/statistical-approaches-to-establishing-bioequivalence-guidance-for-industry-availability
- ICH M13A — Bioequivalence for Immediate-Release Solid Oral Dosage Forms (Oct 2024) — https://www.fda.gov/regulatory-information/search-fda-guidance-documents
- Product-Specific Guidances for Generic Drug Development (FDA database) — https://www.accessdata.fda.gov/scripts/cder/psg/index.cfm
- Common Deficiencies with Bioequivalence Submissions in ANDAs Assessed by FDA (AAPS J review) — https://pmc.ncbi.nlm.nih.gov/articles/PMC3291193/